

Reliability and Transportability of Heart-Disease Classifiers

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Abstract—Machine-learning leaderboards on small public clinical datasets do not establish that fitted probabilities transport. We tested five model families using Cleveland (n=303) for development and Hungary (n=294), Switzerland (n=123) and VA Long Beach (n=200) as locked geographic targets. The primary analysis used eight predictors available across sites; the conventional 13-predictor representation was a labelled sensitivity analysis. Logistic regression, an RBF support-vector machine, random forest, histogram gradient boosting and a multilayer perceptron were selected by five-times-repeated five-fold nested cross-validation with four-fold inner tuning. Frozen pipelines were assessed by ROC-AUC and Brier score, calibration, 5,000-replicate paired uncertainty, dependence- and multiplicity-adjusted comparisons, measured shift, rank stability and controlled neural ablation. Internally, logistic regression achieved AUC 0.869 (95% CI 0.827-0.906) and Brier 0.145 (0.123-0.168); random forest achieved 0.874 and 0.145. No complex-model advantage survived correction (all Holm-adjusted P values were at least 0.964). Hungary transported well (AUC 0.857-0.888), whereas Switzerland and VA fell to 0.747-0.786 and 0.686-0.743. Calibration intercepts were +2.37 to +2.58 in Switzerland and +0.54 to +0.70 in VA, indicating systematic underprediction. The 13-predictor representation improved internal logistic AUC/Brier to 0.909/0.120 but worsened Swiss and VA Brier scores across models; pooled mixed-site CV (AUC 0.862-0.875) concealed these failures. Model complexity did not yield a reliable transport advantage. These historical cohorts support a methodological stress test, not a deployable diagnostic model.

Keywords—calibration; dataset shift; external validation; heart disease; nested cross-validation; transportability

I. INTRODUCTION

High internal discrimination is often treated as evidence that a clinical classifier generalises, yet transport requires more than preserved ranking. A model may retain ROC-AUC while producing probabilities that are systematically too low or too extreme when outcome prevalence, case mix, measurement practice or predictor availability changes. Calibration and proper scoring rules therefore address a different and operationally important question from discrimination (Debray et al. 2015; Steyerberg and Harrell 2016; Van Calster et al. 2019).

The four UCI Heart Disease cohorts provide a compact test of that distinction. They share a processed schema but arose at different historical centres and differ sharply in disease prevalence and missingness (Janosi et al. 1989). The endpoint is prevalent angiographic disease, encoded as num>0; it is not incident cardiovascular risk or prognosis. Cleveland is commonly reused as a self-contained benchmark, which encourages algorithm ranking rather than testing whether the entire source-selected pipeline survives geographic transfer. The cohorts are consequently useful as related geographic or pseudo-external targets, not as contemporary independent clinical validation.

This study asks whether apparent complexity gains are reproducible internally (RQ1), whether they survive dependence-aware uncertainty and multiplicity control (RQ2), how discrimination, probability error, calibration and exploratory net benefit change under frozen transport (RQ3), how those changes correspond descriptively with prevalence, distribution and missingness shift (RQ4), and whether feature rankings and MLP capacity conclusions remain stable across resamples and sites (RQ5). The contribution is an integrated reliability and transportability audit, not a new classifier. The aim is to establish what these benchmark data can and cannot support.

II. RELATED WORK

Routine comparisons of logistic regression, support-vector, tree, boosting and neural models on Cleveland are saturated. More generally, lower-risk clinical comparisons have not demonstrated an average machine-learning advantage over logistic regression, and modern tabular benchmarks show that

flexibility alone is not a guarantee of better generalisation (Christodoulou et al. 2019; Grinsztajn et al. 2022). Recent work has moved beyond single splits: Hasan et al. (2024) addressed cross-dataset discrepancies; Chokphoemphun et al. (2026) used leave-one-country-out evaluation centred on an interpretable decision tree; and Buaphan and Pongphaw (2026) reported cross-country evaluation for random forest rather than every candidate model.

The closest methodological precedents substantially narrow the novelty claim. Gedik (2026) combined leakage-controlled nested cross-validation, uncertainty, calibration and decision curves after pooling the four UCI centres rather than freezing Cleveland fits for site-separated transport. Rathod et al. (2026) combined nested evaluation, calibration, decision curves, explanations and neural ablation on Cleveland but had no external cohort. Paul and Syed Masood (2026) quantified cross-model explanation agreement without frozen geographic transport or resampling- and site-level stability. These studies rule out claims that nested validation, calibration, decision curves, cross-country evaluation, explainability or ablation are new in isolation.

The remaining gap is the controlled integration of these elements: select every candidate under the same repeated source-only protocol, freeze all pipelines, apply each unchanged to three targets, and jointly examine uncertainty, calibration intercept/slope, proper scores, measured shift and explanation stability. Review-level evidence makes this distinction consequential. Cai et al. (2024), whose search ended in July 2021, found no independent validation among 380 reviewed development models and high risk of bias throughout; Liu et al. (2025) found sparse external validation and calibration reporting with extreme heterogeneity. TRIPOD+AI and PROBAST+AI likewise emphasise transparent analysis, calibration, applicability and bounded claims (Collins et al. 2024; Moons et al. 2025). This study is therefore a methodological replication and extension, not a field-wide first.

III. METHODS

A. Data, endpoint and predictor representations

The processed Cleveland, Hungary, Switzerland and VA Long Beach files were parsed as 14 numeric fields. Question

marks were missing. Physiologically impossible zero values were recoded as missing only for resting blood pressure, cholesterol and maximum heart rate; this affected all 123 Swiss cholesterol values, 49 VA cholesterol values and one VA blood-pressure value. No record was removed. Exact repeated rows and negative oldpeak values were retained and flagged. Disease presence was $\text{num} > 0$ (Janosi et al. 1989).

Cleveland was the sole development cohort (Fig. S1). Before model fitting, the primary common representation was fixed from predictor availability and measurement

compatibility alone: age, sex, chest-pain type (cp), resting blood pressure (trestbps), resting ECG (restecg), maximum heart rate (thalach), exercise-induced angina (exang) and ST depression (oldpeak). Each retained predictor had no more than 30% missingness in any cohort; no outcome or model-performance result informed the choice. The conventional 13-predictor sensitivity added cholesterol, fasting blood sugar, ST slope, number of major vessels (ca) and thalassaemia category. It was secondary because the added variables have severe site-specific absence and ca is angiographic and target-proximal (Table I).

TABLE I. COHORTS, OUTCOME PREVALENCE AND PREDICTOR AVAILABILITY

Cohort	n (event/non-event)	Prevalence	Missing cells: 8	Missing cells: 13	Major 13-feature absence
Cleveland	303 (139/164)	45.9%	0 (0.0%)	6 (0.2%)	ca 1.3%; thal 0.7%
Hungary	294 (106/188)	36.1%	4 (0.2%)	782 (20.5%)	slope 64.6%; ca 99.0%; thal 90.5%
Switzerland	123 (115/8)	93.5%	11 (1.1%)	396 (24.8%)	chol 100%; ca 95.9%; thal 42.3%
VA Long Beach	200 (149/51)	74.5%	219 (13.7%)	748 (28.8%)	slope 51.0%; ca 99.0%; thal 83.0%

The eight-predictor representation is primary. The 13-predictor totals include harmonised sentinel recoding. ca records angiographic major-vessel count and is target-proximal.

B. Preprocessing and candidate models

Every learned transformation was fitted within the applicable training fold. Numeric variables were median-imputed and standardised; categorical variables were most-frequent-imputed and one-hot encoded with unknown levels ignored. No missingness indicator, feature selection, class weighting, synthetic resampling or target-site adaptation was used. The secondary classification threshold was fixed at 0.50.

Candidates were L2 logistic regression ($C=0.01-100$), an RBF SVM ($C=0.1-10$; $\gamma=\text{scale}$, 0.01 or 0.1) with sigmoid calibration inside a training-only three-fold procedure containing the full preprocessing pipeline, a 500-tree random forest (depth unrestricted/5/10; feature sampling square-root/0.5; leaf size 1/3/5), histogram boosting (300 iterations; learning rate 0.03/0.1; 7/15/31 leaves; minimum leaf 10/20; L2 0/1; training-only early stopping), and an MLP (8, 32-16 or 64-32-16 hidden units; $\alpha 0.0001/0.001/0.01$; learning rate 0.0003/0.001; ReLU/Adam; training-only early stopping; 1,000 iterations). Scikit-learn 1.8.0 supplied the implementations (Pedregosa et al. 2011).

C. Validation, estimands and inference

Internal evaluation was stratified five-fold outer cross-validation repeated five times, with the same 25 stored splits for all models and both representations. Four-fold inner cross-validation selected hyperparameters by mean ROC-AUC. Final settings were the modal inner selections across the 25 fits; ties used mean inner AUC then a locked parsimony order. Each participant received one held-out probability per repeat. Final pipelines were fitted once on all Cleveland records and applied unchanged to each target. Target outcomes were not used for preprocessing, fitting, tuning, architecture, threshold or feature-performance selection; outcome-free probabilities were hash-locked before labels were merged for evaluation. This procedural lock documents chronology, not blinded validation or external preregistration.

These validation stages answer distinct questions. Repeated nested CV estimates the performance of the Cleveland development procedure: each outer assessment is made on records excluded from both preprocessing and inner-loop selection, so the estimand includes the cost of tuning within this source. Frozen transport instead conditions on one configuration and one fit selected from all Cleveland data. It

asks how that specific source-derived pipeline behaves in each named target when target records are used only for prediction, not to refit, recalibrate or select it; its uncertainty is therefore target-sampling uncertainty conditional on that fit. Mixed-site CV estimates something else again: performance when every training fold contains observations from all four centres. Because training is exposed to the pooled case mix and measurement patterns, it is neither external validation nor an alternative estimate of Cleveland-to-target transport. Its apparently stable average can coexist with poor performance at a particular site because cohort-specific failures are weighted into one pooled score. The three estimates are complementary only if their denominators and conditioning are kept explicit.

Co-primary metrics were ROC-AUC and Brier score (Brier 1950). Secondary metrics were PR-AUC, log loss, accuracy, sensitivity, specificity, precision and F1 at 0.50. Calibration intercept and slope were jointly estimated by logistic recalibration on the logit probability; ideal values are zero and one. Internal estimates averaged the five whole-Cleveland out-of-fold repeat metrics. A participant-cluster bootstrap sampled each participant once while carrying all repeat-specific probabilities and models; target intervals used ordinary participant bootstrap. External intervals therefore condition on the single frozen Cleveland fit and quantify target-sample variability only; they do not propagate source-development or tuning uncertainty. All percentile intervals used 5,000 replicates (Efron 1979).

Formal internal comparisons used paired outer-fold differences, oriented so positive values favoured the candidate. The Nadeau-Bengio repeated-CV variance multiplier was $1/25 + 1/4 = 0.29$ with 24 degrees of freedom; Holm correction covered all eight candidate-by-primary-metric tests (Nadeau and Bengio 2003; Holm 1979). Repeated-fold scores were not treated as independent, DeLong testing was not used on repeated out-of-fold predictions, and non-significance was not interpreted as equivalence. The five analysis hypotheses were locked before execution: no reproducible internal complexity advantage (H1), greater transport deterioration in calibration/proper scores than ranking (H2), no universal winner (H3), lower cross-site than within-source rank agreement (H4), and no MLP capacity benefit persisting across all sites (H5).

Nested CV was also necessary for fair comparison. Hyperparameters were chosen only in the four-fold inner loop; the held-out outer fold then evaluated the whole selection procedure. Using identical stored outer folds for every candidate makes each candidate-versus-logistic difference paired, removing split composition as an avoidable source of comparison noise. Repetition improves stability, but the 25 fold scores are not independent because training sets overlap. A naive t-test would therefore exaggerate effective sample size. The Nadeau-Bengio multiplier inflates the variance using the test-to-train ratio; it is an explicit approximation to this dependence, not proof that the corrected observations become independent. Holm adjustment then controls the family-wise error rate across four candidates and two co-primary metrics, preventing a favourable result from being selected among eight tests without penalty. The participant-cluster bootstrap has a different sampling unit: it resamples Cleveland participants and carries their five OOF predictions for every model together. It preserves person-level pairing and quantifies uncertainty from the observed participant composition, conditional on the realised folds, fitted models and tuning choices. The corrected fold test instead uses variation across retrained outer analyses and is sensitive to training and selection instability. Agreement between them would be reassuring, but disagreement is informative rather than a defect.

D. Transport diagnostics, sensitivities and reproducibility

Reliability curves used five quantile bins. Dataset shift used absolute standardised mean difference and Kolmogorov-Smirnov statistic for observed numeric values, Jensen-Shannon distance for categorical distributions, and absolute missingness differences. These quantities were descriptive and not causal. ROC-AUC permutation importance used 30 repeats for every outer fold/model and frozen target model; agreement used Spearman rho and Kendall tau. A fixed ablation compared 137-, 1,057- and 3,649-parameter MLPs plus stronger L2 on matched splits and seeds. Exploratory decision curves assumed referral for further definitive diagnostic assessment at thresholds 0.10-0.50 and cannot establish clinical utility (Vickers and Elkin 2006; Fisher et al. 2019).

After all primary predictions were locked, a conventional stratified five-fold CV mixing all four sites was run with Cleveland-selected hyperparameters. It could not alter selection. The master seed was 20260818; source and analysis hashes, split-level tuning, fitted pipelines, participant predictions and quality-control reports were retained. All 89 final checks passed and a 104-file hash inventory verified. The only execution deviation was single-process computation after the sandbox denied multiprocessing semaphores; it was logged before the first completed nested fit and changed wall-clock time only.

The reproducibility controls were designed to detect silent execution failures, not merely document filenames. The 89

checks verified raw-file hashes, exact cohort/event and missingness counts, category levels, the approved feature sets, disjoint and exhaustive outer/inner assignments, one OOF probability per participant/repeat/model, expected target rows, finite bounded probabilities, five model families, readable fitted pipelines, final selections, eight corrected tests with valid Holm values, secondary-analysis completeness and figure existence. Prediction locking added a narrower chronological guarantee: the outcome-free target probability files excluded labels and matched their recorded hashes before evaluation merging; it did not imply investigator blinding. Finally, the 104-file inventory records byte counts and SHA-256 hashes for the complete result set, while the submission-level checksum additionally covers code, protocols, data and documentation. Together these checks make accidental substitution, truncation or mixing with the superseded run detectable, but they do not certify scientific validity or contemporary clinical applicability.

IV. RESULTS

A. Internal reliability and corrected comparisons

Under the primary eight predictors, random forest had the highest point AUC, 0.874 (95% CI 0.835-0.910), only 0.006 above logistic regression, 0.869 (0.827-0.906); both had Brier score 0.145. SVM achieved AUC/Brier 0.865/0.148, histogram boosting 0.855/0.154 and MLP 0.826/0.173. No candidate-by-primary-metric comparison survived dependence and multiplicity handling (all Holm-adjusted P values were at least 0.964; Table II). Random forest's corrected AUC difference was +0.004 (-0.012 to 0.021).

The conditional participant bootstrap placed the MLP-minus-logistic AUC difference at -0.043 (-0.061 to -0.025) and its Brier improvement at -0.028 (-0.037 to -0.017), both favouring logistic regression. The corrected fold procedure, which targets training-and-selection variation, did not support a multiplicity-adjusted claim. This contrast is retained because it shows how inferential conclusions depend on the sampling unit; it is neither erased nor recast as equivalence.

The MLP comparison illustrates the distinction. Its negative participant-bootstrap differences show that, for the observed Cleveland participants under the realised repeated-CV fits, logistic regression produced consistently better AUC and Brier results. The corrected-fold analysis asks the stronger question of whether that advantage is reproducible across overlapping retrains after adjustment for eight primary comparisons. With only 25 dependent outer assessments, training and tuning variation widened uncertainty enough that the multiplicity-adjusted claim was not supported. A stable conditional contrast can therefore coexist with weak evidence about the full development procedure. Neither result overrides the other: one describes participant-composition uncertainty conditional on fitted resamples, the other an approximate retraining-level inference. Neither establishes equivalence.

TABLE II. PRIMARY REPEATED NESTED PERFORMANCE AND CORRECTED COMPARISONS

Model	ROC-AUC (95% CI)	Brier (95% CI)	Corrected delta AUC (95% CI); P	Corrected delta Brier (95% CI); P
Logistic regression	0.869 (0.827-0.906)	0.145 (0.123-0.168)	Reference	Reference
SVM	0.865 (0.823-0.902)	0.148 (0.127-0.170)	-0.005 (-0.016,+0.006); 1.000	-0.003 (-0.010,+0.004); 1.000
Random forest	0.874 (0.835-0.910)	0.145 (0.127-0.165)	+0.004 (-0.012,+0.021); 1.000	-0.000 (-0.010,+0.010); 1.000
Histogram boosting	0.855 (0.815-0.892)	0.154 (0.132-0.178)	-0.015 (-0.037,+0.008); 1.000	-0.009 (-0.023,+0.004); 1.000
MLP	0.826 (0.783-0.866)	0.173 (0.157-0.189)	-0.033 (-0.081,+0.016); 1.000	-0.028 (-0.063,+0.008); 0.964

Positive corrected deltas favour the candidate; P values are Holm-adjusted across eight co-primary tests. Participant-cluster CIs condition on the fitted repeated-CV scheme.

B. Frozen geographic transport

Hungary transported unexpectedly well: AUC ranged from 0.857 (MLP) to 0.888 (random forest), and Brier from 0.129 to 0.154. Switzerland AUC ranged from 0.747 to 0.786 and Brier from 0.159 to 0.180; its intervals were broad because 115 of 123 participants had disease. The corresponding PR-AUCs near 0.98 largely reflected that extreme prevalence and should not be read as universally strong classification. In VA,

AUC ranged from 0.686 to 0.743 and Brier from 0.170 to 0.189 (Fig. 1; Table III).

No model led every site and dimension. Random forest led internal and Hungary point AUC; MLP had the lowest Swiss Brier; SVM had the highest VA AUC; histogram boosting had the lowest VA Brier. Target paired intervals generally overlapped. H3 was therefore supported descriptively, while the well-transporting Hungary result showed that a visible marginal shift need not produce performance loss.

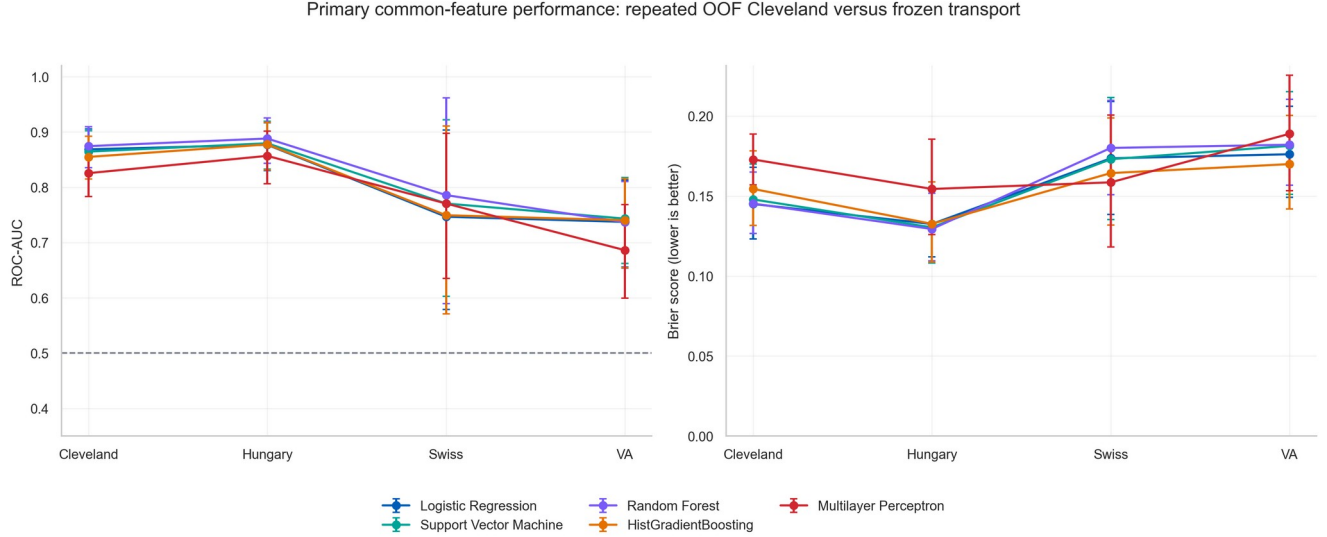


Fig. 1. Primary eight-predictor performance under repeated out-of-fold Cleveland evaluation and frozen geographic transport. Points are model estimates; bars are 95% participant-bootstrap intervals. Lower Brier score is better. The lines connect sites only to aid model tracing, not to imply a temporal ordering.

TABLE III. LOCKED PRIMARY EXTERNAL PERFORMANCE AND CALIBRATION-IN-THE-LARGE

Cohort	Model	ROC-AUC (95% CI)	Brier (95% CI)	Calibration intercept (95% CI)
Hungary	Logistic regression	0.877 (0.830-0.917)	0.133 (0.112-0.155)	-0.29 (-0.61-0.02)
Hungary	SVM	0.880 (0.833-0.919)	0.130 (0.108-0.155)	-0.33 (-0.66-0.01)
Hungary	Random forest	0.888 (0.844-0.925)	0.129 (0.110-0.152)	-0.33 (-0.65-0.01)
Hungary	Histogram boosting	0.878 (0.831-0.917)	0.133 (0.109-0.159)	-0.36 (-0.69-0.05)
Hungary	MLP	0.857 (0.806-0.902)	0.154 (0.126-0.186)	-0.65 (-1.00-0.36)
Switzerland	Logistic regression	0.747 (0.579-0.903)	0.174 (0.138-0.209)	2.50 (1.87-3.89)
Switzerland	SVM	0.771 (0.603-0.922)	0.173 (0.135-0.212)	2.51 (1.87-4.01)
Switzerland	Random forest	0.786 (0.590-0.962)	0.180 (0.151-0.210)	2.58 (1.93-4.26)
Switzerland	Histogram boosting	0.749 (0.571-0.911)	0.164 (0.132-0.199)	2.43 (1.79-3.81)
Switzerland	MLP	0.771 (0.635-0.897)	0.159 (0.118-0.201)	2.37 (1.74-3.70)
VA Long Beach	Logistic regression	0.737 (0.656-0.813)	0.176 (0.149-0.206)	0.60 (0.25-0.98)
VA Long Beach	SVM	0.743 (0.663-0.818)	0.181 (0.151-0.215)	0.68 (0.33-1.05)
VA Long Beach	Random forest	0.737 (0.654-0.811)	0.182 (0.157-0.210)	0.69 (0.36-1.06)
VA Long Beach	Histogram boosting	0.740 (0.654-0.815)	0.170 (0.142-0.200)	0.54 (0.18-0.91)
VA Long Beach	MLP	0.686 (0.600-0.769)	0.189 (0.153-0.226)	0.57 (0.19-0.96)

Intercept zero is ideal; positive values indicate aggregate underprediction. No target recalibration was performed. Switzerland has only eight non-events, so its calibration and discrimination intervals are imprecise.

C. Calibration failure and measured shift

Cleveland calibration intercepts were near zero; logistic regression's intercept/slope were -0.001/1.050. Hungary intercepts were negative (-0.289 to -0.651), with MLP slope 0.703. Every Swiss intercept was strongly positive (+2.371 to +2.584), consistent with systematic underprediction in a

93.5%-prevalence cohort; slopes were highly uncertain. VA intercepts were +0.541 to +0.695 and slopes 0.445-0.761, showing aggregate underprediction and excessive probability spread for several models (Fig. 2). H2 was only partly supported because discrimination also deteriorated and not every proper score worsened relative to the weak internal MLP.

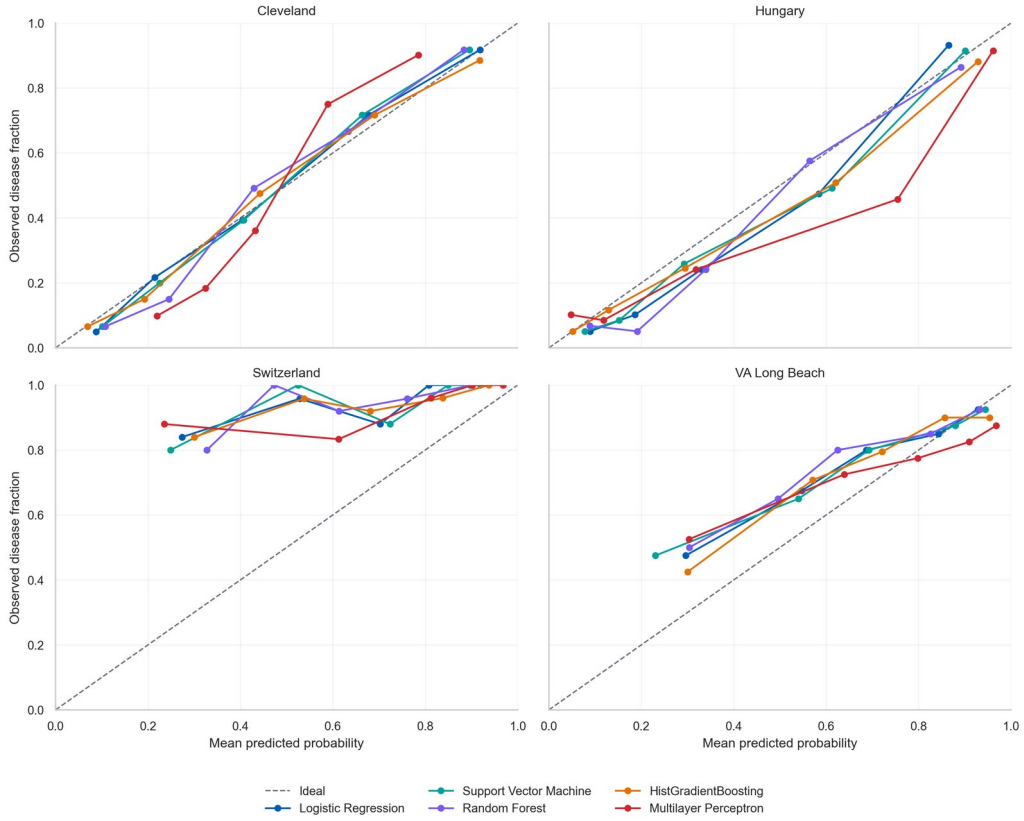


Fig. 2. Five-bin reliability curves for repeated out-of-fold Cleveland probabilities and frozen target predictions. For Cleveland, plotted probabilities are each participant's mean across five repeats; numerical calibration metrics average the five repeat-specific metrics. The dashed diagonal is ideal calibration. Sparse Swiss non-events make curve shape unstable, but the consistent vertical displacement agrees with the large positive calibration intercepts.

Prevalence differed from Cleveland by -9.8 percentage points in Hungary, +47.6 in Switzerland and +28.6 in VA. Among primary predictors, maximum heart rate had absolute SMD 1.18 in Switzerland and 1.19 in VA; age shifted 0.78 in Hungary and 0.57 in VA; and resting-ECG Jensen-Shannon distance reached 0.49-0.53 in Switzerland/VA. The conventional predictors added structural absence: Swiss cholesterol 100% and ca 95.9%; VA ca 99.0%, thalassaemia 83.0% and slope 51.0% (Fig. 3). These differences coincided with transport failure but do not isolate its cause.

D. Stability, ablation and sensitivity analyses

Within-Cleveland mean Spearman rank agreement ranged from 0.55 for MLP to 0.78 for logistic regression/SVM. Cleveland-to-Hungary agreement was 0.74-0.95 and VA 0.52-0.83, but Switzerland fell to 0.02-0.29 for all models. Chest-pain type led the mean Cleveland permutation ranking for four models and oldpeak for MLP. H4 was only partly supported: geographic instability was concentrated in Switzerland, while Hungary and VA often preserved source rankings. These are predictive, model-conditional associations, not causal risk factors. The complete rank-agreement matrix is shown in Fig. S2.

The 137-parameter MLP performed poorly internally (AUC/Brier 0.699/0.224); 1,057 parameters improved to 0.813/0.181 and 3,649 to 0.839/0.169. Stronger L2 produced 0.836/0.172. Capacity therefore mattered, but the medium network had higher VA AUC than the large variants (0.731 versus about 0.719), so no capacity conclusion persisted across all sites and metrics. H5 was supported only in that narrower, predeclared sense. The complete controlled ablation is shown in Fig. S3.

The conventional 13-predictor representation looked stronger internally: logistic AUC/Brier improved from 0.869/0.145 to 0.909/0.120. Raw Brier score is prevalence-dependent, so cross-cohort differences are descriptive rather than pure degradation measures. Within-site contrasts hold prevalence fixed: the 13 predictors did not yield consistent target AUC gains and worsened probability error where added measurements were structurally absent. Swiss Brier was 0.219-0.353 and VA 0.242-0.332, versus 0.159-0.180 and 0.170-0.189 with eight predictors. Separately, post-lock mixed-site CV yielded AUC 0.862-0.875 and Brier 0.141-0.147, obscuring the weaker Swiss/VA locked transport (Table IV).

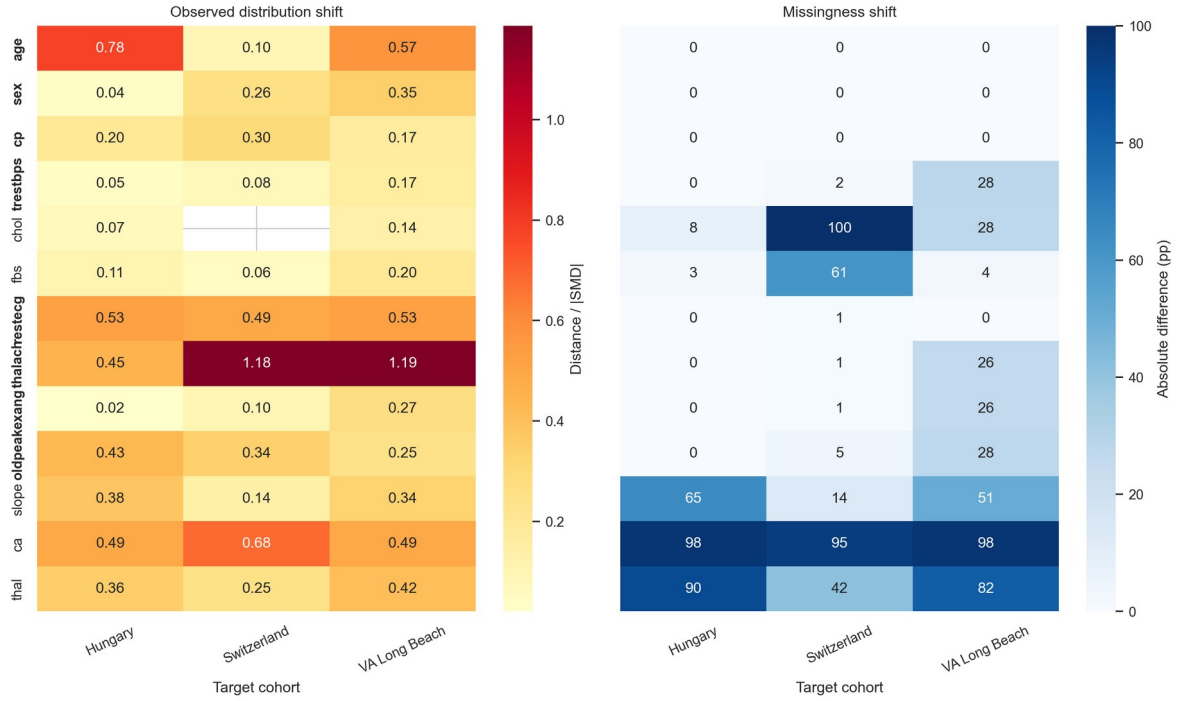


Fig. 3. Cleveland-to-target observed distribution and missingness shift. Numeric cells are absolute SMD and categorical cells Jensen-Shannon distance; primary predictors are bold. Blank Swiss cholesterol distribution reflects complete post-harmonisation absence. Missingness is the absolute percentage-point difference from Cleveland.

TABLE IV. REPRESENTATION AND POOLED-VALIDATION SENSITIVITIES

Model	13-feature Cleveland AUC/Brier	Hungary Brier 8->13	Swiss Brier 8->13	VA Brier 8->13	Mixed-site 8-feature AUC/Brier
Logistic regression	0.909/0.120	0.133->0.127	0.174->0.259	0.176->0.267	0.875/0.142
SVM	0.906/0.121	0.130->0.124	0.173->0.258	0.181->0.260	0.875/0.141
Random forest	0.908/0.124	0.129->0.142	0.180->0.277	0.182->0.279	0.875/0.142
Histogram boosting	0.885/0.136	0.133->0.159	0.164->0.353	0.170->0.332	0.862/0.147
MLP	0.860/0.158	0.154->0.123	0.159->0.219	0.189->0.242	0.874/0.141

The 13-feature analysis is sensitivity only. Mixed-site values are conventional five-fold CV with Cleveland-selected hyperparameters, run after primary predictions were locked; they are not estimates of site-separated transport.

Exploratory decision curves exceeded treat-all and treat-none across Cleveland and almost all Hungary thresholds. Treat-all dominated nearly all Swiss thresholds because prevalence was 93.5%; VA differences were small and inconsistent. These curves diagnose threshold/prevalence interaction and do not establish a useful clinical action. Results were consistent with H1 in its prespecified operational sense: no corrected complexity advantage was detected. This does not establish equivalence or exclude effects smaller than the study could resolve. H3 was supported descriptively; H2, H4 and H5 were only partly supported. All negative and contradictory findings were retained. The full threshold curves are shown in Fig. S4.

V. DISCUSSION

The central result is not a winning classifier but a limit on what internal model comparison can establish. Random forest's point AUC advantage over logistic regression was 0.006, Brier scores were essentially identical, and no corrected primary comparison supported reproducible superiority. The MLP was materially weaker under participant-conditional inference, yet the dependence-corrected fold test was uncertain. This is consistent with the broader observation that added flexibility on small tabular datasets can increase selection and optimisation burden without dependable benefit (Christodoulou et al. 2019; Grinsztajn et al. 2022). It does not prove model equivalence.

The more consequential failure was calibration transport. Moderate Swiss and VA AUCs showed that ranking information survived to some degree, while large positive intercepts showed that the absolute probabilities did not. ROC-AUC is invariant to monotone rescaling; calibration is not. The high-prevalence cohorts therefore produced the clinically awkward combination of above-chance ranking, systematic underprediction and threshold behaviour often no better than treating everyone as positive. This claim rests chiefly on intercept/slope and reliability evidence, not raw cross-site Brier ordering. No threshold or recalibration was learned from a target, so this is a stress test of frozen probabilities rather than model updating.

ROC-AUC and calibration address different properties. AUC is the probability that a randomly selected diseased participant is ranked above a non-diseased participant; any strictly increasing transformation of predictions leaves that ranking unchanged. A transported model can therefore retain useful ordering while assigning probabilities that are systematically wrong. The calibration intercept measures average log-odds misalignment after allowing for slope: a positive intercept indicates underprediction overall, as in Switzerland and VA. A slope below one indicates predictions that vary too extremely, whereas a slope above one indicates insufficient spread, although sparse outcome groups make both estimates unstable. Brier score is a proper score for the

full probability forecast, combining calibration with separation or refinement, but its attainable range and baseline difficulty depend on prevalence. That is why raw Brier values were not treated as prevalence-free cross-site degradation. Within one cohort, the eight-versus-13-feature Brier comparison holds prevalence fixed; across cohorts, intercept, slope and reliability curves carry the clearest evidence that probability transport failed even where AUC remained above chance.

Measured shift provides a plausible but non-causal account. Prevalence, maximum heart rate, age and resting-ECG distributions differed, while missingness represented measurement practice as well as absent patient information. Median or modal imputation keeps a pipeline executable; it cannot reconstruct a variable that a site did not collect. The 13-predictor sensitivity makes this visible: *ca*, *thalassaemia*, slope and cholesterol strengthened internal Cleveland estimates but became absent or nearly absent in targets, and probability error deteriorated. Target-proximal angiographic *ca* also weakens the claim that this conventional representation corresponds to a practical pre-diagnostic predictor set.

Structural missingness is more serious than occasional item non-response. When almost an entire target site lacks *ca*, *thal*, slope or cholesterol, source-trained median or modal imputation supplies a conventional source value to nearly everyone. This preserves a numerical input shape but contributes little patient-specific information and can shift the meaning of the fitted decision function; no imputer can reconstruct measurements that were never collected. The 13-feature model can thus exploit Cleveland-only signals internally yet lose them systematically at transport. Its within-site comparison with the eight-feature pipeline diagnoses the practical consequence of deploying that representation under the target's measurement process, not a causal effect attributable to any single missing variable. The common eight-feature set reduced this measurement mismatch while retaining the conventional representation as an explicit stress test.

Feature stability was equally context-dependent. Moderate-to-high repeat and Hungary/VA agreement could have encouraged a universal explanation claim, but Swiss correlations near zero refuted it. Stable source importance therefore does not guarantee explanatory transport, especially when distributions and measurement processes differ. Permutation importance is conditional on the fitted model, correlated predictors and sampled population; it should be interpreted as predictive reliance rather than a causal clinical ranking (Fisher et al. 2019).

The pooled-CV sensitivity clarifies why many benchmark studies can report stable AUC near 0.87 while missing transport failure. Mixing sites lets every training fold observe the broad multi-site distribution and averages over the very heterogeneity that a source-to-target question seeks to expose. Its estimand is legitimate but different. The site-separated design is stronger for the present question because each target remained outside preprocessing, selection and fitting. Likewise, the 13-predictor result shows why an internally better representation need not be the more transportable one.

Together these findings support model selection by proportional complexity: extra capacity should earn its optimisation, explanation and maintenance cost through reproducible improvements in discrimination and probability quality across intended settings. Avoiding unjustified complexity can also reduce needless computational and

energy demand, although this small benchmark was not a lifecycle assessment. The results support joint reporting of discrimination, calibration and proper scores rather than choosing a model from AUC alone. The present data do not justify a clinical operating threshold, a universal feature explanation or deployment. The publishable contribution is the failure analysis and its reproducible audit trail.

VI. LIMITATIONS AND FUTURE WORK

All four cohorts are small, historical, clinically selected members of the same UCI collection. Switzerland has only eight non-events, making discrimination and calibration-slope estimates especially unstable. The binary endpoint collapses disease severity; predictor timing relative to diagnostic work-up is uncertain; recruitment dates, treatment, sampling frames and contemporary outcome verification are absent. These targets are geographic/pseudo-external cohorts, not independent prospective validation. Confidence intervals quantify sampling uncertainty under assumptions but cannot make the samples representative.

The study compared five broad families rather than every modern tabular architecture. Bootstrap intervals condition on observed participants; the corrected repeated-fold test is approximate; calibration coefficients can be unstable in sparse outcome groups; and multiple secondary analyses are exploratory. Cross-site rank coefficients are point summaries over only eight predictors, while the controlled MLP configurations cannot isolate parameter count from stochastic optimisation or early stopping. No equivalence margin, causal shift model, adequately powered subgroup/fairness analysis or validated clinical action was defined. Decision curves are diagnostic demonstrations only. The single-process deviation affected runtime rather than the statistical design.

A stronger next study should define the target population, prediction time and clinical action before modelling; recruit contemporary independently governed sites with standardised predictor timing; retain site identifiers; and use internal-external cross-validation or hierarchical meta-analysis to estimate heterogeneity. Missingness mechanisms should be studied prospectively, with multiple-imputation or missing-not-at-random sensitivity where defensible. Any target recalibration must be learned on target development data and evaluated on a separate sample. Subgroup performance, fairness, monitoring and impact evaluation require adequate sample sizes and governance (Efthimiou et al. 2024; Collins et al. 2024; Moons et al. 2025).

VII. CONCLUSION

Across four UCI Heart Disease cohorts, validation design and cohort context mattered more than model class. No complex model produced a corrected internal advantage over logistic regression with the primary common predictors. Hungary transported well, but Switzerland and VA showed weaker discrimination and systematic calibration failure; feature rankings and neural-capacity conclusions were site-dependent. The conventional 13-predictor and mixed-site pooled-CV sensitivities demonstrated how attractive internal or pooled summaries can conceal structural missingness and target-specific probability error. Reliable heart-disease modelling therefore requires site-separated transport assessment, calibration and bounded claims before complexity or headline AUC is promoted as superiority.

DECLARATIONS

Ethics and participants: This secondary analysis used public, de-identified historical data and recruited no

participants. No individual identities were available. Patient and public involvement: None.

Data and code availability: Data are available from the UCI Machine Learning Repository (doi: 10.24432/C52P4X) under CC BY 4.0. The locked protocol, code, environment versions, seeds, hashes, fitted pipelines, predictions and quality-control reports accompany the public repository at <https://github.com/aarongulzar/heart-disease-transportability-audit>. This release is a software artefact and technical report; it has not been peer reviewed.

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SUPPLEMENTARY FIGURES

These figures complete the visual audit trail for the locked design, explanation stability, neural-capacity sensitivity and exploratory threshold analysis. They are descriptive or sensitivity evidence and do not broaden the clinical claims of the main report.

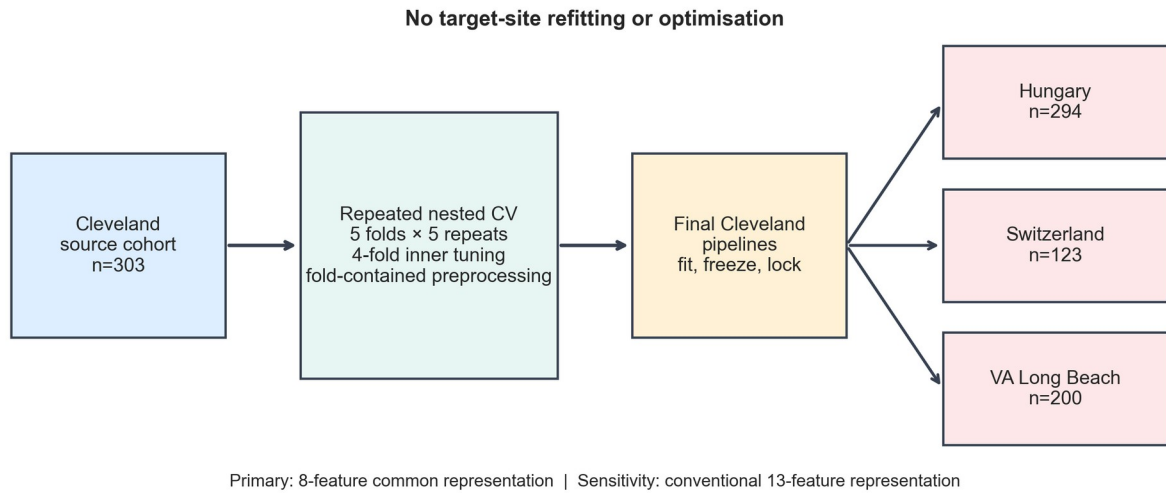


Fig. S1. Locked source-to-target design. Cleveland alone supplied preprocessing, tuning and final fitting; no target-site refitting or optimisation was performed. The shared eight-predictor representation was primary and the conventional 13-predictor representation was a labelled sensitivity analysis.

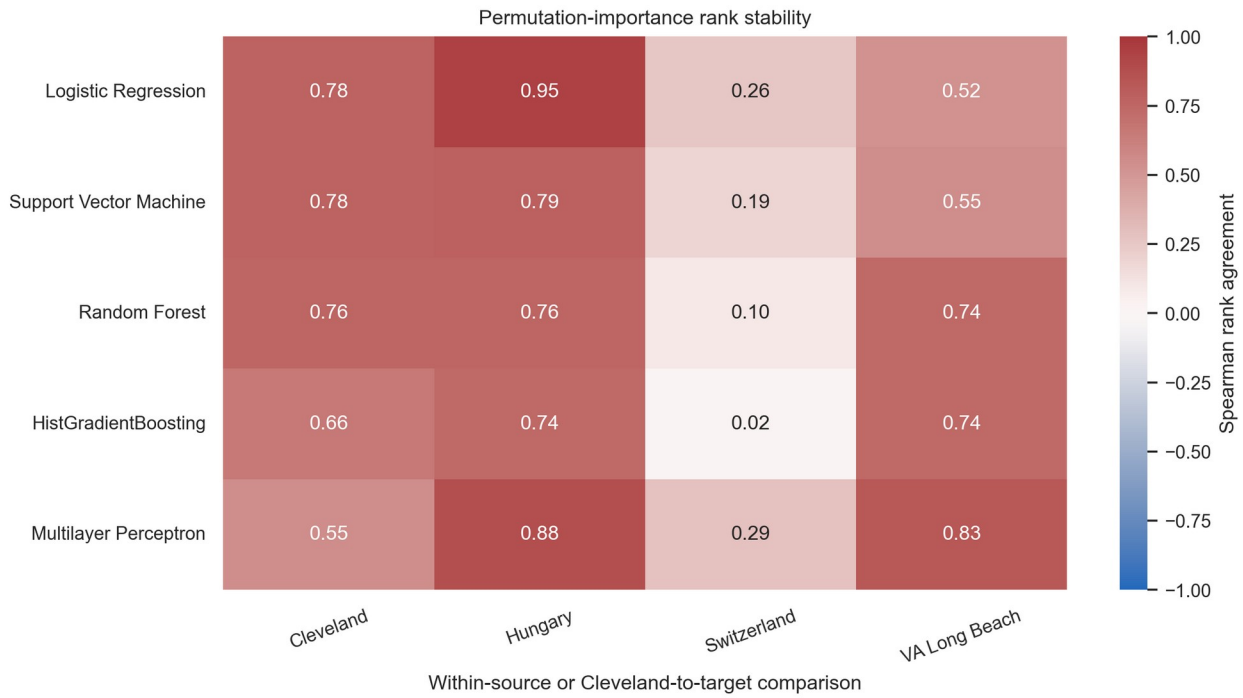


Fig. S2. Permutation-importance rank stability. Cells report mean within-Cleveland or Cleveland-to-target Spearman agreement across the eight primary predictors. Switzerland showed near-zero agreement for every model, bounding claims of universal explanation stability.

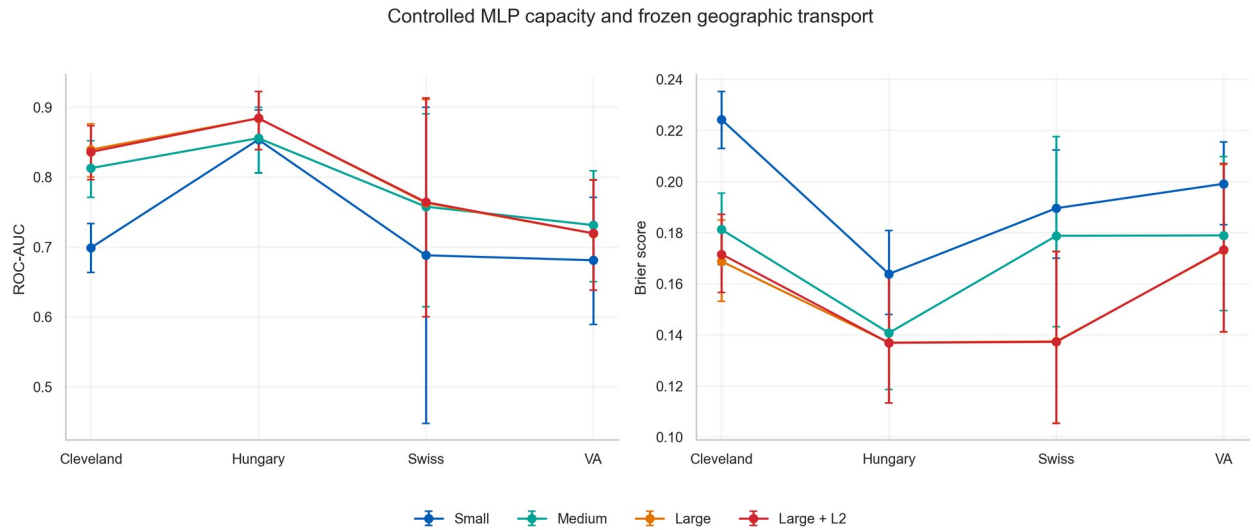


Fig. S3. Controlled MLP capacity ablation under matched splits and seeds. Greater capacity improved the weak small network internally, but no capacity ordering persisted across all transported cohorts and metrics. Bars are 95% participant-bootstrap intervals.

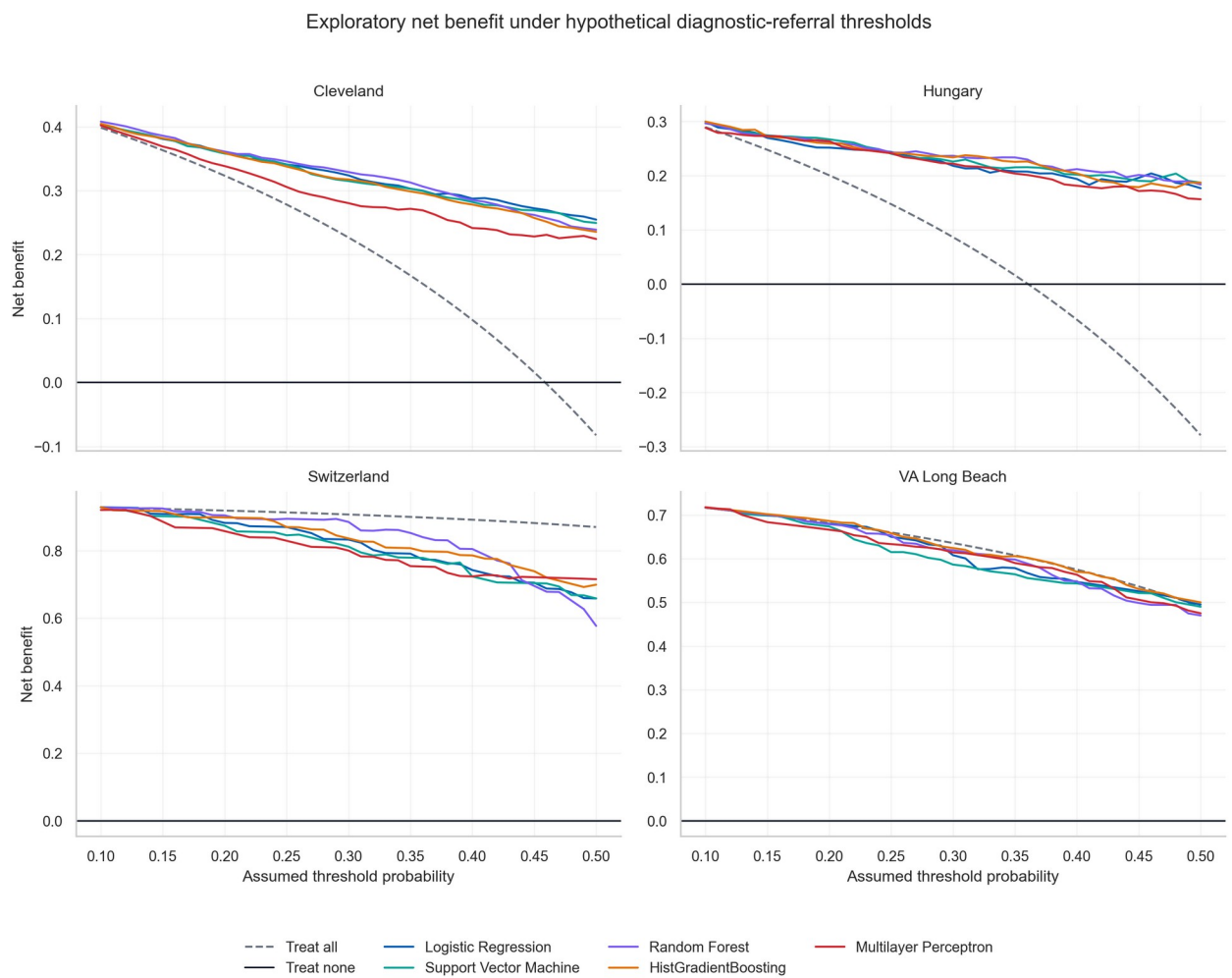


Fig. S4. Exploratory decision curves at hypothetical diagnostic-referral thresholds from 0.10 to 0.50. These curves diagnose threshold-prevalence interaction; they do not establish a validated operating threshold, clinical utility or a recommended intervention.